

Cyber Security Task 2 (2026)

Phishing Email Detection & Awareness System

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INTRODUCTION

Phishing attacks are one of the most common cyber threats used by attackers to steal sensitive information such as passwords, banking details, OTPs, and personal data.

Attackers use fake emails and malicious links to trick users into revealing confidential information. These attacks mainly target human behavior through fear, urgency, and fake notifications.

This project analyzes sample phishing emails, identifies phishing indicators, classifies email risks, and provides awareness guidelines to help users recognize and avoid phishing attacks.

OBJECTIVE

Objective:

To analyze phishing email samples, identify phishing indicators, classify risks, and create awareness guidelines for users to improve cybersecurity awareness.

TOOLS USED

- Microsoft Word
- Browser
- Google Message Header Analyzer
- MXToolbox Email Header Analyzer
- Screenshot Tool

EMAIL SAMPLE 1 — PHISHING EMAIL

Email

Urgent: Your Bank Account Will Be Suspended

Dear Customer,

We detected suspicious activity in your bank account.

To avoid temporary suspension, verify your account immediately using the secure link below:

[http://secure-bank-verification\[.\]com](http://secure-bank-verification[.]com)

Failure to verify within 24 hours may result in account suspension.

Regards,
Bank Security Team

Analysis

Email Classification: PHISHING

Risk Level: HIGH

Phishing Indicators

-
1. Generic Greeting
 - Uses “Dear Customer” instead of recipient name
 2. Urgent Language
 - Creates fear using suspension warning
 3. Suspicious Link
 - Fake verification domain
 4. Insecure URL
 - Uses HTTP instead of HTTPS

Explanation

This phishing email attempts to scare users into clicking a fake verification link. The attacker uses urgency and fear tactics to trick the victim into sharing banking credentials.

EMAIL SAMPLE 2 — SUSPICIOUS EMAIL

Email

Employee Payroll Update Required

Hello Employee,

The HR department has updated payroll records for this month.

Please download and review the attached payroll file immediately.

Attachment:

Payroll_Update.zip

Thank you,
HR Department

Analysis

Email Classification: SUSPICIOUS

Risk Level: MEDIUM

Indicators

1. ZIP Attachment
 - May contain malware
2. Generic Greeting
 - Uses "Hello Employee"
3. Urgent Language
 - Requests immediate action
4. Unexpected Attachment
 - Unusual payroll file

Explanation

This email contains an unexpected ZIP attachment that may contain malicious files or malware. Users should avoid downloading unknown attachments without verification.

EMAIL SAMPLE 3 — SAFE EMAIL

Email

Your Microsoft Account Security Update

Hello Deekshitha,

Your Microsoft account security settings were successfully updated on May 18, 2026.

If you made this change, no further action is required.

You can review your account activity securely at:
<https://account.microsoft.com/security>

Thank you,
Microsoft Security Team

Analysis

Email Classification: SAFE

Risk Level: LOW

Indicators

1. Personalized Greeting
 - Uses recipient name
2. Secure HTTPS Link
 - Official secure link
3. Official Domain
 - microsoft.com
4. Professional Tone
 - No fear or urgency tactics

Explanation

This email appears legitimate because it uses an official Microsoft domain, secure HTTPS links, personalized greeting, and professional communication style.

PREVENTION GUIDELINES

- Never click suspicious or unknown links
- Verify sender email domains carefully
- Avoid opening unexpected attachments
- Use Multi-Factor Authentication (MFA)
- Report suspicious emails to the IT/security team
- Check for HTTPS and official domains before logging in
- Avoid sharing passwords or OTPs through email

EMPLOYEE DO'S & DON'TS

DO'S:

- Verify sender domains
- Use MFA
- Report phishing emails
- Check official websites

DON'TS:

- Don't click suspicious links
- Don't share OTPs
- Don't open unknown attachments
- Don't trust urgent threats immediately

CONCLUSION

This project demonstrated how phishing attacks use fake domains, malicious links, urgency, and suspicious attachments to trick users into revealing sensitive information.

By analyzing phishing indicators and understanding common attack methods, users can improve cybersecurity awareness and reduce the risk of phishing attacks.